Week 4

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1 Retrieval-Augmented Generation

1.1 Introduction

- RAG is an architecture for optimizing the performance of LLMs by connecting them to external knowledge bases; help LLMs deliver more relevant responses at a higher quality.
- Prompt → LLM → User Query → Information → LLM → User
- Use cases for RAG:
  - **Question Answering**: can retrieve the latest and most relevant information from sources before generating answer.
  - **Research Augmentation**: research accuracy is high as it combines historical data with recent trends to inform research findings.
  - **Content Generation**: improves factual accuracy and overall credibility of content by including credible citations.
  - **Domain-specific and proprietary data integration**: can retrieve information from internal cases without embedding it with training data and shared to public domain.

1.2 Benefits

- AI hallucination is a phenomenon in which an LLM detects patterns or objects that do not exist or are not visible to humans, leading to inaccurate or meaningless outputs.

- To confirm that organizations receive current, complete, and accurate responses is to retrain or fine-tune the model with updated data; but it is costly and impractical, so RAG is a solution to keep expanding its knowledge without retaining constantly.
- Without retraining, businesses can scale AI applications while reducing costs and resource demands.
- Allows to integrate with online APIs, social media feeds, and search engines to get information from external sources and stay up-to-date with market trends, news, and customer sentiment.

1.3 RAG Workflow

- It consists of steps that combine information extracted from knowledge sources to produce more informed, precise, and relevant responses.
- **Prompt Creation**: submits a query that requires a response acting like a trigger that prompts RAG to get user input, determine the intent and scope of the information, and retrieve the relevant information.
- **Information Search**: searches external knowledge sources (research papers, books, company databases, APIs, web articles) which allows the response to the user be beyond the training dataset and most recent.
- **Information Retrieval**: filters and extracts the most relevant and credible information, making sure that only the high-quality data aligned with the user's query is selected.
- **Prompt Augmentation**: enhances the original query by integrating the retrieved information by providing the model with additional context to generate a more accurate and detailed answer aligned with the user's intent.
- **Response Generation**: uses enhanced information to generate a clear, accurate, and actionable response which is given to the user.